

Dexter Fernandes

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<https://github.com/Dexter-Fernandes>

Machine Learning Engineer

Machine Learning Engineer with 3+ years of experience delivering low-latency vision solutions for Tier 1 companies, including a construction firm in Texas and a UK oil and gas company. Proven track record of collaborating with multinational clients to deploy AI-driven vision solutions in fast-paced dynamic environments. Experienced in applying state-of-the-art AI research while leading testing, QA, and monitoring processes.

WORK EXPERIENCE

Esbaar • Muscat, Oman

01/2023 - 08/2023

AI Engineer

- Developed time-series machine learning models for the classification of on-shore oil rig operations to increase worker safety and reduce supervision required during 24/7 operation, achieving a 23% higher F1 score than human experts.
- Built an automated sensor-data cleansing pipeline (3.6 M data points) that cut model roll out from 3 days to less than 24 hours.
- Led strategic MLOps initiatives to continuously enhance experimentation and deployment workflows through bayesian hyperparameter tuning of time-series and vision models.
- Spearheaded the end-to-end development and production rollout of YOLOv5 models, to elevate rig worker and equipment detection precision by 15% in 3 months.
- Refactored a legacy monolithic system into a Kafka-driven microservice architecture, containerising each service with Docker to enhance failure isolation and improve MTBF by 35% within four months
- Designed a dynamic memory management solution to address recurring crashes in a live video streaming system, reducing crash frequency by 67%.
- Enhanced code quality and documentation standards by leading formal peer code reviews and quality-assurance activities within each sprint cycle, ensuring best practices.
- Developed a tool for collecting raw sensor data and applying automated augmentation and labeling workflows to boost dataset diversity by 40% and cut manual annotation effort by 60%.

LivNSense Technologies Pvt. Ltd. • Bengaluru, India

05/2020 - 12/2022

Lead Software Engineer - AI • Full-time

- Led a 4-engineer team on ViCAS to design, train, and deploy deep learning models for real-time object detection, pose estimation, and tracking on edge hardware to reduce false positives by 33% while achieving sub-50 ms inference latency within 6 months.
- Deployed quantised OpenVINO vision models over Microsoft Azure to rugged Intel VPU edge devices for on-site inference in harsh industrial environments.
- Repurposed legacy object detection models for automated labelling pipeline workflows, reducing manual annotation effort by 80% and accelerating dataset curation.
- Engineered synthetic training data pipelines using OpenCV and NumPy, reducing edge-case model development time from 6 days to 2 days for high-risk production environments.
- Optimized neural network architectures using quantization-aware training, structured pruning, and layer fusion thus reducing model size by 50%.
- Implemented rigorous testing and validation frameworks based on simulated adverse conditions, to ensure model accuracy and stability, reducing performance variance by 25% under challenging lighting, occlusion, and noise scenarios.

EDUCATION

Master of Science in Robotics - Merit

University of Bristol • Bristol, England

09/2023 - 09/2024

- Developed and implemented a novel real-time SLAM solution by integrating LiDAR-based mapping (RTAB-Map) with gamma-ray detection on an Unmanned Ground Vehicle (UGV).
- Conducted extensive tests to evaluate mapping accuracy using Velodyne VLP-16 LiDAR, achieving reliable performance in diverse environments, including indoor high-detail and outdoor feature-scarce areas.
- Optimised RTAB-Map parameters specifically for enhanced performance in varying terrain conditions, significantly improving the accuracy of localisation and mapping in challenging outdoor settings.
- Conducted simulations for 2D and 3D SLAM, as well as waypoint navigation of the Clearpath Husky robot in Gazebo, using ROS2 Humble to validate and tune navigation algorithms in both indoor and outdoor scenarios
- Validated the system through rigorous real-world experimentation, including the deployment of the robot in complex industrial environments, effectively navigating obstacles and identifying radioactive sources in real-time.

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

07/2016 - 05/2020

CERTIFICATIONS

Deep Learning Specialisation

deeplearning.ai

07/2019

AWARDS & SCHOLARSHIPS

Best Speaker award

ABC Toastmasters club

01/2020

SKILLS

- **Perception & Robotics:** GStreamer, LiDAR SLAM, Nav2, OpenCV, OpenVINO, ROS1 (Noetic), ROS2 (Humble/Foxy/Jazzy), Vision Transformer (ViT), YOLOv5/v8
- **Backend & Cloud / Edge:** Apache Kafka, AWS, FastAPI, Intel VPUs, Microsoft Azure, NVIDIA Jetson AGX, PostgreSQL
- **DevOps & MLOps:** Docker, DVC (Data Version Control), Git, Linux, MLFlow, Optuna
- **Programming Languages:** C++, Python
- **Testing & Monitoring:** gtest, pytest